Agenda
- Decision Trees → Pure makes / overfits
- ensemble Techniques
  - → Type → Bagging
  - Random forest
    - bag → lot of trees → DT
  - Bias — variance
  - Hyper parameters
  - sk-learn
  - Time / space complexity

Ensemble Techniques

→ group of things → multiple models

$m_1, m_2, \ldots, m_k$ → base models

combine → M → Powerful

1. Bagging (Random forest)
2. Boosting
3. stacking (kaggle) Gigle DS competitions
4. Cascading

Bagging → (Bootstrap) (Aggregation)

Sampling with replacement

Train Data D_m → $D_m^{(1)}$, $D_m^{(2)}$, ..., $D_m^{(k)}$ → m_1, m_2, ..., m_k → aggregate → M

k times → k models

m → sample size (bootstrapping)

n (total data points) (Train)

$m \ll n$, $1000 \ll 10{,}000$

- mean
- median
- mode

(n-m) removed

model → variance (1) → overfit

low

k=3

k=1 (very high variance)

Bagging → sampling (replacement) → reduce variance

base models → DT → high variance

bagging → reduce variance

base model: high variance + low bias + bagging: reduce variance

= low variance & low bias

m_1, m_2, m_3 → Pattern Attack

Random Forest

Sampling (Sampling) Column (Column) Sample (Sample)

D_m → $D_m^{(1)}$, $D_m^{(2)}$, ..., $D_m^{(k)}$ → single bag → m_1, m_2, ..., m_k → avg → M

k times

RF = DT + RS + CS/feature sampling (base model)

$D^{(1)}$: $n' \ll n$, $d' \ll d$

$n = 10000$, $n' = 200, 1000$

$d = 100$, $d' = 20, 25$

Q1 will RF >>> DT most of the time.

Q2 when should I go for DT over RF?

get data is very simple

Time & space complexity

Train 1 DT → $(n \log n \times d)$

Prediction time 1 DT → $O(\text{depth})$

RF → Prediction Time: depth × k → n-estimators (base learners)

Training time → $k \to (n d \log n)$

$k \to k d' n' \log n'$

m_1, m_2, m_3, ..., m_k → same time (parallel) parallel processing

d_1, d_2, d_3, ..., d_{10}

$d_1 \times 0.1$, $d_2 \times 0.2$, $d_1 \times 0.1$, $d_1 \times 0.25$

$0.1 + 0.2 + 0.25$

10

15

7, 7, 7 → n-estimators

	5	5	7	8	10
100	500	—	—	—	—
200	—	—	X	—	—
300	—	—	—	3.26	—
400	—	—	—	—	—

n=100, n=200, n=300, n=400

type of, optimal → 3D of optimal

1. depth of base learners (high variance)
2. n-estimators

3. CSR (column sampling rate) (max - floating) → overfit
   RSR (row sampling rate) (max - samples)
4. n-estimators → underfit